Waypoint ANALYTICAL

2790 Whitten Road, Memphis, TN 38133
Main 901.213.2400 ⁰ Fax 901.213.2440
www.waypointanalytical.com

7/30/2024

Charm Industrial
Lina Zhu
2575 Marin St
San Francisco, CA, 94124

Ref: Analytical Testing
Lab Report Number: 24-200-0020
Client Project Description: Basco 6 SWD MRV
Formation Fluid

Dear Lina Zhu:
Waypoint Analytical, LLC. received sample(s) on 7/18/2024 for the analyses presented in the following report.

The above referenced project has been analyzed per your instructions. The analyses were performed in accordance with the applicable analytical method. Where the laboratory was not responsible for the sampling stage (refer to the chain of custody) results apply to the sample as received.

The analytical data has been validated using standard quality control measures performed as required by the analytical method. Quality Assurance, method validations, instrumentation maintenance and calibration for all parameters (NELAP and non-NELAP) were performed in accordance with guidelines established by the USEPA (including 40 CFR 136 Method Update Rule May 2021) and NELAC unless otherwise indicated. Any parameter for which the laboratory is not officially NELAP accredited is indicated by a '~' symbol. These are not included in the scope because NELAP accreditation is either not available or has not been applied for. Additional certifications may be held/are available for parameters, where NELAP accreditation is not required or applicable. A full list of certifications is available upon request.

Certain parameters (chlorine, pH, dissolved oxygen, sulfite...) are required to be analyzed within 15 minutes of sampling. Usually, but not always, any field parameter analyzed at the laboratory is outside of this holding time. Refer to sample analysis time for confirmation of holding time compliance.

The results are shown on the attached Report of Analysis(s). Results for solid matrices are reported on an as-received basis unless otherwise indicated. This report shall not be reproduced except in full and relates only to the samples included in this report.

Please do not hesitate to contact me or client services if you have any questions or need additional information.

Sincerely,

Connie Cook

Connie Cook
Project Manager

*Laboratory's liability in any claim relating to analyses performed shall be limited to, at laboratory's option, repeating the analysis in question at laboratory's expense, or the refund of the charges paid for performance of said analysis.*

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# Certification Summary

**Laboratory ID: WP MTN: Waypoint Analytical, LLC., Memphis, TN**

| State | Program | Lab ID | Expiration Date |
|---|---|---|---|
| Alabama | State Program | 40750 | 02/28/2025 |
| Arkansas | State Program | 88-0650 | 02/07/2025 |
| California | State Program | 2904 | 06/30/2024 |
| Florida | State Program - NELAP | E871157 | 06/30/2025 |
| Georgia | State Program | C044 | 11/14/2025 |
| Georgia | State Program | 04015 | 06/30/2024 |
| Illinois | State Program - NELAP | 200078 | 10/31/2024 |
| Kentucky | State Program | 90047 | 12/31/2024 |
| Kentucky | State Program | 80215 | 06/30/2025 |
| Kentucky | State Program | KY90047 | 12/31/2024 |
| Louisiana | State Program - NELAP | LA037 | 12/31/2024 |
| Louisiana | State Program - NELAP | 04015 | 06/30/2025 |
| Mississippi | State Program | MS | 11/14/2025 |
| North Carolina | State Program | 47701 | 07/31/2024 |
| North Carolina | State Program | 415 | 12/31/2024 |
| Pennsylvania | State Program - NELAP | 68-03195 | 05/31/2025 |
| South Carolina | State Program | 84002 | 06/30/2024 |
| Tennessee | State Program | 02027 | 11/14/2025 |
| Texas | State Program - NELAP | T104704180 | 09/30/2024 |
| Virginia | State Program | 00106 | 06/30/2025 |
| Virginia | State Program - NELAP | 460181 | 09/14/2024 |

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## Sample Summary Table

**Report Number:** **24-200-0020**

**Client Project Description:** **Basco 6 SWD MRV Formation Fluid**

| Lab No | Client Sample ID | Matrix | Date Collected | Date Received |
|---|---|---|---|---|
| 98021 | Formation Fluid | Aqueous | 07/17/2024 08:00 | 07/18/2024 |

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Client: Charm Industrial
Project: Basco 6 SWD MRV
Lab Report Number: 24-200-0020
Date: 7/29/2024

**CASE NARRATIVE**

**Total Metals Method EPA-200.7**
Sample 98021 (Formation Fluid)
QC Batch No: L763834/L763545
One or more Internal Standards are outside method acceptance criteria. Re-analysis and/or sample dilutions are required.

Sample 98021 (Formation Fluid)
Analyte: Potassium
QC Batch No: L763834/L763545
This analyte was analyzed at a dilution due to IEC interference/matrix interference.

Sample 98021 (Formation Fluid)
Analyte: Sodium
QC Batch No: L763834/L763545
This analyte was analyzed at a dilution due to IEC interference/matrix interference.

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32714
Charm Industrial
Lina Zhu
2575 Marin St
San Francisco , CA 94124

Project Basco 6 SWD MRV

Information : Formation Fluid

Report Date : 07/30/2024
Received : 07/18/2024

Connie Cook

Report Number : **24-200-0020**

***REPORT OF ANALYSIS***

Connie Cook
Project Manager

Lab No : **98021**
Sample ID : **Formation Fluid**

Matrix: **Aqueous**
Sampled: **7/17/2024 8:00**

| Test | Results | Units | MQL | DF | Date / Time Analyzed | By | Analytical Method |
|---|---|---|---|---|---|---|---|
| Specific Gravity | **1.01** | g/mL | | 1 | 07/29/24 15:00 | HBT | ASTM D-1298-85 |
| Alkalinity (as CaCO3) | **502** | mg/L | 20 | 1 | 07/29/24 11:15 | HMR | 2320B-2011 |
| Bicarbonate | **502** | mg/L | 20 | 1 | 07/29/24 11:15 | HMR | 2320B-2011 |
| Carbonate | <2 | mg/L | 2 | 1 | 07/29/24 11:15 | HMR | 2320B-2011 |
| Chloride | **24200** | mg/L | 400 | 1000 | 07/23/24 14:12 | HMQ | EPA-300.0 |
| Conductivity | **67400** | µS/cm | 0.010 | 1 | 07/29/24 15:56 | HHS | 2510B-2011 |
| pH | **7.7** | s.u. | | 1 | 07/19/24 13:14 | ABB | 4500H+B-2011 |
| Total Dissolved Solids | **23300** | mg/L | 2500 | 1 | 07/19/24 13:46 | CAH | 2540C-2015 |
| Sulfate | **20.6** | mg/L | 10.0 | 10 | 07/24/24 15:47 | HMQ | EPA-300.0 |
| Barium | **33.4** | mg/L | 0.100 | 10 | 07/26/24 12:27 | TJS | EPA-200.7 |
| Calcium | **1250** | mg/L | 1.00 | 10 | 07/26/24 12:27 | TJS | EPA-200.7 |
| Iron | **1160** | mg/L | 1.00 | 10 | 07/26/24 12:27 | TJS | EPA-200.7 |
| Magnesium | **112** | mg/L | 1.00 | 10 | 07/26/24 12:27 | TJS | EPA-200.7 |
| Potassium | **144** | mg/L | 2.00 | 20 | 07/29/24 13:06 | TJS | EPA-200.7 |
| Sodium | **12600** | mg/L | 250 | 500 | 07/29/24 13:11 | TJS | EPA-200.7 |

**Qualifiers/ Definitions** DF Dilution Factor MQL Method Quantitation Limit

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# Quality Control Data

**Client ID:** **Charm Industrial**
**Project Description:** **Basco 6 SWD MRV**
**Report No:** **24-200-0020**

**QC Analytical Batch:** L764108
**Analysis Method:** 2320B-2011
**Analysis Description:** Alkalinity

**Laboratory Control Sample** LCS

| Parameter | Units | Spike Conc. | LCS Result | LCS %Rec | % Rec Limits |
|---|---|---|---|---|---|
| Alkalinity (as CaCO3) | mg/L | 472 | 458 | 97.0 | 85-115 |

**Duplicate** L 99690-DUP

| Parameter | Units | Result | DUP Result | RPD | Max RPD | Analyzed |
|---|---|---|---|---|---|---|
| Alkalinity (as CaCO3) | mg/L | 122 | 122 | 0.0 | 10 | 07/29/24 11:15 |
| Bicarbonate | mg/L | 122 | 122 | 0.0 | 10 | 07/29/24 11:15 |
| Carbonate | mg/L | < 2 | <2 | 0.0 | 10 | 07/29/24 11:15 |

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# Quality Control Data

**Client ID:** **Charm Industrial**
**Project Description:** **Basco 6 SWD MRV**
**Report No:** **24-200-0020**

**QC Analytical Batch:** L764221
**Analysis Method:** 2510B-2011
**Analysis Description:** Conductivity/Resistivity

**Duplicate** L 93957-DUP

| Parameter | Units | Result | DUP Result | RPD | Max RPD | Analyzed |
|---|---|---|---|---|---|---|
| Conductivity | µS/cm | 274 | 273 | 0.3 | 10.0 | 07/29/24 15:56 |

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# Quality Control Data

**Client ID:** **Charm Industrial**
**Project Description:** **Basco 6 SWD MRV**
**Report No:** **24-200-0020**

**QC Analytical Batch:** L762564
**Analysis Method:** 2540C-2015
**Analysis Description:** Total Dissolved Solids

**Lab Reagent Blank** LRB Matrix: AQU
Associated Lab Samples: 98021

| Parameter | Units | Blank Result | MQL | Analyzed |
|---|---|---|---|---|
| Total Dissolved Solids | mg/L | < 50.0 | 50.0 | 07/19/24 13:46 |

**Laboratory Control Sample** LCS

| Parameter | Units | Spike Conc. | LCS Result | LCS %Rec | % Rec Limits |
|---|---|---|---|---|---|
| Total Dissolved Solids | mg/L | 250 | 256 | 102 | 90-110 |

**Duplicate** L 97657-DUP

| Parameter | Units | Result | DUP Result | RPD | Max RPD | Analyzed |
|---|---|---|---|---|---|---|
| Total Dissolved Solids | mg/L | 814 | 807 | 0.8 | 10.0 | 07/19/24 13:46 |

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# Quality Control Data

**Client ID:** Charm Industrial
**Project Description:** Basco 6 SWD MRV
**Report No:** 24-200-0020

**QC Prep:** L763545
**QC Prep Batch Method:** EPA-200.7 (PREP)
**QC Analytical Batch(es):** L763834
**Analysis Method:** EPA-200.7
**Analysis Description:** Total Metals

**Lab Reagent Blank** LRB-L763545 Matrix: AQU
Associated Lab Samples: 98021

| Parameter | Units | Blank Result | MQL | Analyzed |
|---|---|---|---|---|
| Barium | mg/L | < 0.0100 | 0.0100 | 07/26/24 00:33 |
| Calcium | mg/L | 0.130 | 0.100 | 07/26/24 00:33 |
| Iron | mg/L | < 0.100 | 0.100 | 07/26/24 00:33 |
| Magnesium | mg/L | < 0.100 | 0.100 | 07/26/24 00:33 |
| Potassium | mg/L | < 0.100 | 0.100 | 07/26/24 00:33 |
| Sodium | mg/L | < 0.500 | 0.500 | 07/26/24 00:33 |

**Laboratory Control Sample** LCS-L763545

| Parameter | Units | Spike Conc. | LCS Result | LCS %Rec | % Rec Limits |
|---|---|---|---|---|---|
| Barium | mg/L | 1.00 | 1.00 | 100 | 85-115 |
| Calcium | mg/L | 10.0 | 10.6 | 106 | 85-115 |
| Iron | mg/L | 10.0 | 10.1 | 101 | 85-115 |
| Magnesium | mg/L | 10.0 | 9.85 | 99.0 | 85-115 |
| Potassium | mg/L | 1.00 | 1.01 | 101 | 85-115 |
| Sodium | mg/L | 10.0 | 9.61 | 96.0 | 85-115 |

**Matrix Spike & Matrix Spike Duplicate** L 89494-MS-L763545 L 89494-MSD-L763545

| Parameter | Units | Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS %Rec | MSD %Rec | %Rec Limits | RPD | Max RPD |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Barium | mg/L | 0.0218 | 1.00 | 1.00 | 1.02 | 1.01 | 100 | 99.0 | 70-130 | 0.9 | 20.0 |
| Calcium | mg/L | 25.6 | 10.0 | 10.0 | 36.1 | 35.6 | 105 | 100 | 70-130 | 1.3 | 20.0 |
| Iron | mg/L | < 0.100 | 10.0 | 10.0 | 10.2 | 10.0 | 101 | 99.0 | 70-130 | 1.9 | 20.0 |
| Magnesium | mg/L | 5.19 | 10.0 | 10.0 | 15.2 | 15.0 | 100 | 98.0 | 70-130 | 1.3 | 20.0 |
| Potassium | mg/L | 1.58 | 1.00 | 1.00 | 2.62 | 2.58 | 104 | 100 | 70-130 | 1.5 | 20.0 |

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# Quality Control Data

**Client ID:** **Charm Industrial**
**Project Description:** **Basco 6 SWD MRV**
**Report No:** **24-200-0020**

**QC Prep:** L763545
**QC Prep Batch Method:** EPA-200.7 (PREP)
**QC Analytical Batch(es):** L763834
**Analysis Method:** EPA-200.7
**Analysis Description:** Total Metals

**Matrix Spike & Matrix Spike Duplicate** L 89494-MS-L763545 L 89494-MSD-L763545

| Parameter | Units | Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS %Rec | MSD %Rec | %Rec Limits | RPD | Max RPD |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sodium | mg/L | 4.79 | 10.0 | 10.0 | 14.6 | 14.5 | 98.0 | 97.0 | 70-130 | 0.6 | 20.0 |

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# Quality Control Data

**Client ID:** **Charm Industrial**
**Project Description:** **Basco 6 SWD MRV**
**Report No:** **24-200-0020**

**QC Prep:** L763336
**QC Prep Batch Method:** EPA-300.0 (PREP)
**QC Analytical Batch(es):** L763392
**Analysis Method:** EPA-300.0
**Analysis Description:** Anions by Ion Chromatography

**Lab Reagent Blank** LRB-L763336 Matrix: AQU
Associated Lab Samples: 98021

| Parameter | Units | Blank Result | MQL | Analyzed |
|---|---|---|---|---|
| Chloride | mg/L | < 0.400 | 0.400 | 07/23/24 12:55 |

**Laboratory Control Sample & LCSD** LCS-L763336 LCSD-L763336

| Parameter | Units | Spike Conc. | LCS Result | LCSD Result | LCS %Rec | LCSD % Rec | % Rec Limits | RPD | Max RPD |
|---|---|---|---|---|---|---|---|---|---|
| Chloride | mg/L | 50.0 | 49.1 | 50.3 | 98.0 | 101 | 90-110 | 2.4 | 20.0 |

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# Quality Control Data

**Client ID:** **Charm Industrial**
**Project Description:** **Basco 6 SWD MRV**
**Report No:** **24-200-0020**

**QC Prep:** L763581
**QC Prep Batch Method:** EPA-300.0 (PREP)
**QC Analytical Batch(es):** L763671
**Analysis Method:** EPA-300.0
**Analysis Description:** Anions by Ion Chromatography

**Lab Reagent Blank** LRB-L763581 Matrix: AQU
Associated Lab Samples: 98021

| Parameter | Units | Blank Result | MQL | Analyzed |
|---|---|---|---|---|
| Sulfate | mg/L | < 1.00 | 1.00 | 07/24/24 10:29 |

**Laboratory Control Sample & LCSD** LCS-L763581 LCSD-L763581

| Parameter | Units | Spike Conc. | LCS Result | LCSD Result | LCS %Rec | LCSD % Rec | % Rec Limits | RPD | Max RPD |
|---|---|---|---|---|---|---|---|---|---|
| Sulfate | mg/L | 62.5 | 64.4 | 64.3 | 103 | 103 | 90-110 | 0.1 | 20.0 |

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## Shipment Receipt Form

Customer Number: **32714**
Customer Name: **Charm Industrial**
Report Number: **24-200-0020**

### Shipping Method

| | | | | |
|---|---|---|---|---|
| (●) Fed Ex | ( ) US Postal | ( ) Lab | ( ) Other : | |
| ( ) UPS | ( ) Client | ( ) Courier | Thermometer ID: | N/A |

| | | | |
|---|---|---|---|
| Shipping container/cooler uncompromised? | (●) Yes | ( ) No | |
| Number of coolers/boxes received | 1 | | |
| Custody seals intact on shipping container/cooler? | ( ) Yes | ( ) No | (●) Not Present |
| Custody seals intact on sample bottles? | ( ) Yes | ( ) No | (●) Not Present |
| Chain of Custody (COC) present? | (●) Yes | ( ) No | |
| COC agrees with sample label(s)? | (●) Yes | ( ) No | |
| COC properly completed | (●) Yes | ( ) No | |
| Samples in proper containers? | (●) Yes | ( ) No | |
| Sample containers intact? | (●) Yes | ( ) No | |
| Sufficient sample volume for indicated test(s)? | (●) Yes | ( ) No | |
| All samples received within holding time? | (●) Yes | ( ) No | |
| Cooler temperature in compliance? | ( ) Yes | ( ) No | (●) Not Present |
| Cooler/Samples arrived at the laboratory on ice. Samples were considered acceptable as cooling process had begun. | ( ) Yes | (●) No | |
| Water - Sample containers properly preserved | (●) Yes | ( ) No | ( ) N/A |
| Water - VOA vials free of headspace | ( ) Yes | ( ) No | (●) N/A |
| Trip Blanks received with VOAs | ( ) Yes | ( ) No | (●) N/A |
| Soil VOA method 5035 – compliance criteria met | ( ) Yes | ( ) No | (●) N/A |

☐ High concentration container (48 hr) ☐ Low concentration EnCore samplers (48 hr)
☐ High concentration pre-weighed (methanol -14 d) ☐ Low conc pre-weighed vials (Sod Bis -14 d)

| | | |
|---|---|---|
| Special precautions or instructions included? | ( ) Yes | (●) No |

Comments: Sample was received outside of temperature compliance.

Signature: Makayla Weaver Date & Time: 07/18/2024 09:45:25

| Client Name/Address | Client Project Manager/Contact | Billing Information | For Laboratory Use Only |
|---|---|---|---|
| Charm<br>2575 Mann St SFCA 94124 | JAnderson<br>661-282-0138 | | |

| Project Description | Project/Site Location (City/State) | | Method of Shipment | Matrix Key |
|---|---|---|---|---|
| Basco 6 SWD MRV<br>Formation fluid | Basco 6 SWD | ☐ RUSH – Additional charges apply<br>☐ Special Detection Limit(s)<br>Date Results Needed | ☒ Fed Ex ☐ UPS ☐ USPS<br>☐ Courier ☐ Client Drop Off<br>Other | WW – Wastewater GW – Groundwater<br>DW – Drinking Water S – Soil /Solid O – Oil<br>P - Product M - Misc |

| Project Number | Project Manager Phone # | Project Manager Email | Purchase Order Number | Site/Facility ID # |
|---|---|---|---|---|
| | 661-282-0138 | jennifer.anderson@charmindustrial.com | | |

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Memphis, TN 38133
(901) 213-2400

Unless noted, all containers per Table II of 40 CFR Part 136.

| | |
|---|---|
| A | Cool < 10C Na2S2O3 (Micro Only) |
| B | Cool <= 6C |
| C | H2SO4 pH<2 |
| D | None Required |
| E | NaOH pH>10 |
| F | HNO3 pH<2 |
| G | HCL pH<2 |
| H | H3PO4 pH<2 |
| I | Cool <= 6C NA2S2O3 |

| Date | Time | Sample Identification | Number of Containers | Matrix (Refer to Key) | (G)rab or (C)omposite | Required Analysis / Preservative: pH, density, chlorides, conductivity, Sodium, Calcium, Mg, Fe, Barium, K, HCO3, Sulfate, Carbonate, TDS | Comments/Notes |
|---|---|---|---|---|---|---|---|
| 07/17 | 0800 | Formation fluid | 1 | WW | C | | give a shout w/ concerns |
| | | | | | | | |
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24-200-0020
32714
07-18-2024
09:44:47
Charm Industrial
Basco 6 SWD MRV

| For Laboratory Use Only | | | Sampled by (Name – Print) | Client Remarks/Comments |
|---|---|---|---|---|
| Ice: Y / N (Y) | Custody Seals: Y / N (Y) | Lab Comments: Sample was received out of temp. compliance. Not for compliance per J Anderson. (W) 7/18/24 | Jennifer Anderson | |
| Blank/Cooler Temp: N/A | | | | |

| Relinquished by: (SIGNATURE) | Date Time | Received by: (SIGNATURE) | Date Time |
|---|---|---|---|
| J Anderson | 7/17 0800 | | |
| | | | |
| | | | 7/18/24 0902 |